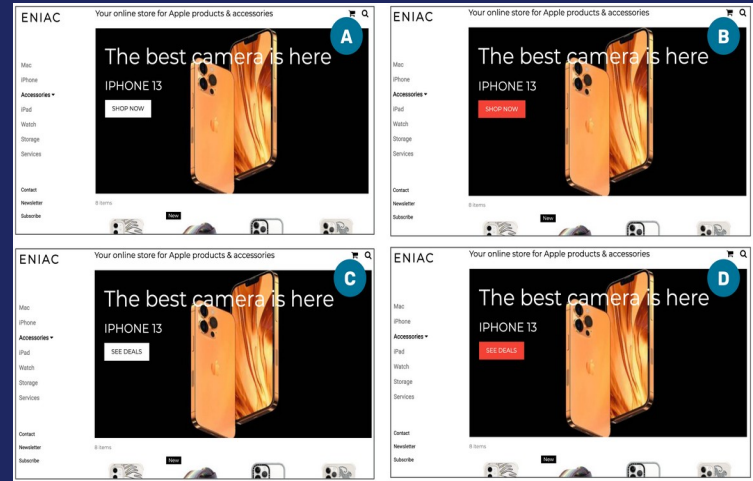


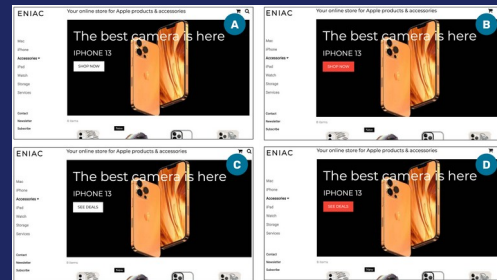
A/B Testing

Multivariate Testing



Multivariate Testing

A 2x2 Factorial Design for the SHOP NOW Button



⚠ Often called "A/B Testing" — but technically Multivariate (MVT)

Original Copy

"SEE DEALS" Copy

Original
Color

Version A

Control

Version C

New copy only

Red
Color

Version B

Red color only

Version D

Both changes

Key Design Questions Before Launching



A/B or Multivariate?

With 2 variables (color × copy) and 4 combinations, this is an MVT — specifically a 2×2 factorial design, not a simple A/B test



What metric to track?

Primary KPI = Click-Through Rate (CTR) — a discrete/binomial metric.
Secondary: bounce rate, revenue per visitor



How to split traffic?

Random assignment (server-side or CDN-level) ensures equal, unbiased exposure per variant — essential for causal inference



Guarding against chance?

Statistical significance ($p < 0.05$) & minimum detectable effect set before launch. Chi-squared test fits discrete CTR data



How long to run?

Run until sample size is met (typically 1–4 weeks). Avoid stopping early — let the data settle to avoid false positives



Other elements to test?

Hero section, search & pricing UX have higher traffic — but test one section at a time to isolate learnings

Conclusions & Recommendations

01

This is MVT, not A/B

Testing color × copy simultaneously across 4 versions is a 2×2 factorial design. It reveals not just which variable helps, but whether they interact — something a simple A/B test misses entirely

02

Comparison → Causal Relation

The test compares independent groups (comparison), but random assignment lets us infer causation: red button → higher CTR → more sales. That causal power is what makes MVT the gold standard

03

Plan Before You Peek

Set sample size, significance threshold ($\alpha=0.05$) and run duration before launch. Checking results early inflates false-positive risk — especially with 4 variants running simultaneously

 MVT is iterative — each experiment informs the next hypothesis. Start small, scale what works.

Thank you for your attention

NOT EVERYTHING
THAT COUNT
COUNTS AND NOT
EVERYTHING
THAT COUNTS
CAN BE COUNTED



ASIN : B0B5P92HF6